

[Menu](#)

Advanced Features

▼ Relevant source files

`hclient/framecontent/infoPage.php``hclient/framecontent/initPageLogin.php``hclient/widgets/cms/cmsTemplate.php``hclient/widgets/cms/editCMS2.js``hclient/widgets/cms/editCMS_ElementCfg.html``hclient/widgets/cms/editCMS_ElementCfg.js``hclient/widgets/cms/editCMS_SelectElement.html``hclient/widgets/cms/editCMS_SelectElement.js``hclient/widgets/cms/editCMS_WidgetCfg.html``hclient/widgets/cms/editCMS_WidgetCfg.js``hclient/widgets/cms/websiteRecord.php``hclient/widgets/cms/websiteScriptAndStyles.php``hclient/widgets/search/searchInput.js``hclient/widgets/viewers/app_storymap.js``hclient/widgets/viewers/app_timemap.js``hclient/widgets/viewers/recordListExt.js``viewers/map/mapDocument.js``viewers/map/mapDraw.php``viewers/map/mapLayer2.js``viewers/map/mapManager.js``viewers/map/mapPreview.php`

`viewers/map/mapping.js`

Purpose and Scope

This page documents the specialized features of Heurist that extend its core capabilities for advanced use cases. These features include an integrated mapping system, a content management system (CMS) for website generation, and data analysis/visualization tools. While the core Heurist system focuses on data management and basic search capabilities, these advanced features allow users to present, visualize, and interact with their data in sophisticated ways.

For information about core system architecture, see [System Architecture](#). For details about the user interface framework that supports these features, see [UI Framework](#). For basic data import and export functionality, see [Data Management](#).

Mapping System

The mapping system provides robust geospatial visualization capabilities, integrating with Heurist's core record management to display geographic data on interactive maps.

Architecture and Components

The mapping system is built on the Leaflet JavaScript library and consists of several interrelated components:

Mapping System Architecture

Sources: `viewers/map/mapping.js` `viewers/map/mapManager.js` `viewers/map/mapDocument.js`
`viewers/map/mapLayer2.js`

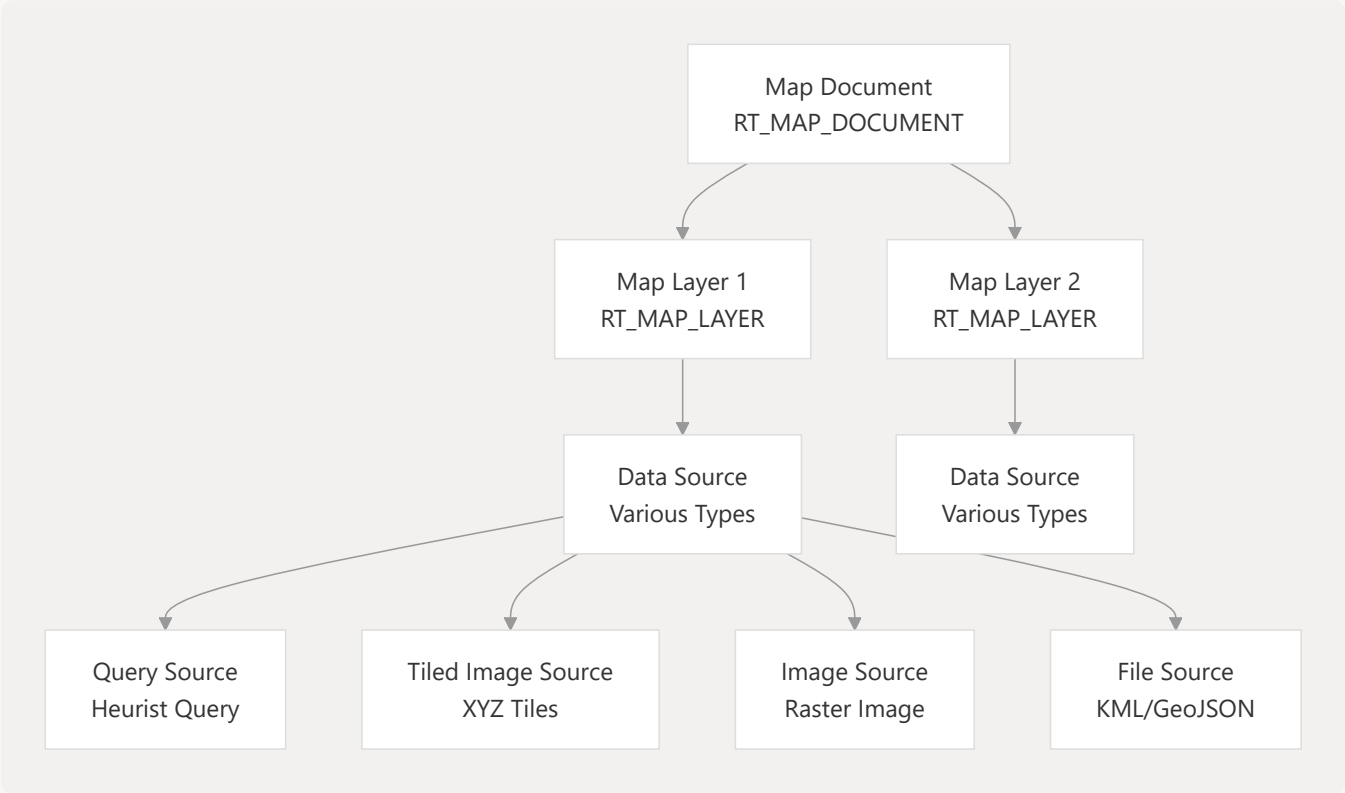
Key Components

- **mapping.js**: The main widget that provides core mapping functionality through a jQuery UI widget called `heurist.mapping`. It handles map initialization, event management, and integration with other components.
- **mapManager.js**: Manages the map legend, base maps, and layer visibility through the `HMapManager` class.
- **mapDocument.js**: Handles map documents and their content through the `hMapDocument` class. Map documents are Heurist records that define collections of layers.

- **mapLayer2.js:** Interface between Heurist data sources and Leaflet layers through the `HMapLayer2` class. Supports various layer types like GeoJSON, tile layers, and image overlays.

Map Document Structure

Map documents organize geographic data in a hierarchical structure:



Sources: `viewers/map/mapDocument.js` 58-98

Supported Layer Types

The mapping system supports multiple layer types:

Layer Type	Description	Source Class
GeoJSON	Vector data from Heurist records or files	HMapLayer2
Tile Layers	Base maps and other tiled services	L.tileLayer
Image Overlays	Georeferenced raster images	L.imageOverlay
KML/GeoJSON Files	External geographic data files	HMapLayer2
Query Sources	Dynamic data from Heurist queries	HMapLayer2

Sources: `viewers/map/mapLayer2.js` 144-993

Map-Timeline Integration

A unique feature of the Heurist mapping system is its integration with timeline visualization, allowing users to explore both spatial and temporal dimensions of their data. The timeline component works with the map to filter data by time periods.

```
// Code snippet from mapping.js showing timeline integration
updateTimelineData: function(layer_id, layer_data) {
    // Add/replace timeline data for a specific layer
    // This updates the timeline without reloading the entire dataset
}
```

Sources: viewers/map/mapping.js 60-62

Digitizing Tools

The mapping system includes tools for creating and editing geographic features:

Tool	Purpose	Methods
Draw Points	Create point features	drawLoadSimplePoints()
Draw Lines	Create linear features	drawLoadWKT()
Draw Polygons	Create area features	drawLoadJson()
Edit Features	Modify existing features	setFeatureSelection()

Sources: viewers/map/mapping.js 80-92 viewers/map/mapDraw.php

Base Maps and Symbology

The system provides access to numerous base maps through the Leaflet Providers plugin and supports custom symbology for vector features:

```
// From mapping.js - defining available base maps
basemap_providers: [
    {name: 'OpenStreetMap'},
    {name: 'OpenTopoMap'},
    {name: 'Esri.WorldStreetMap'},
    {name: 'Esri.WorldImagery'},
    // Many more options...
]
```

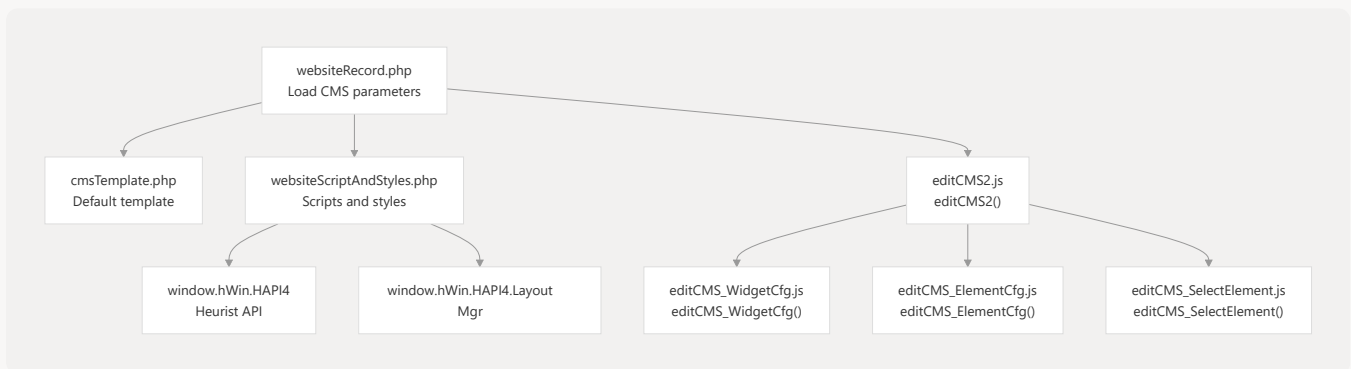
Sources: `viewers/map/mapping.js` 254-293

CMS Website Generator

Heurist's CMS capabilities enable users to create websites directly from their database content, with minimal technical knowledge required.

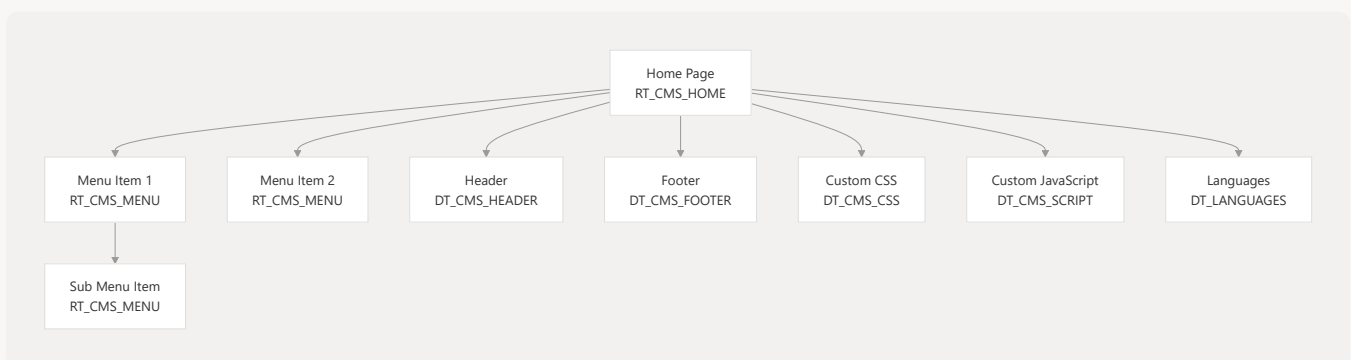
Architecture

The CMS system comprises PHP server-side components and JavaScript client-side editing tools:

Sources: `hclient/widgets/cms/websiteRecord.php` `hclient/widgets/cms/websiteScriptAndStyles.php``hclient/widgets/cms/cmsTemplate.php``hclient/widgets/cms/editCMS2.js`

Website Structure

Websites are structured as a hierarchy of records:

Sources: `hclient/widgets/cms/websiteRecord.php` 33-48`hclient/widgets/cms/websiteRecord.php` 104-103

CMS Editor

The CMS editor provides a WYSIWYG interface for designing and editing website pages:

Component	Purpose	File
editCMS2.js	Main editor framework	hclient/widgets/cms/editCMS2.js
editCMS_WidgetCfg.js	Widget configuration dialog	hclient/widgets/cms/editCMS_WidgetCfg.js
editCMS_ElementCfg.js	CSS and element properties editor	hclient/widgets/cms/editCMS_ElementCfg.js
editCMS_SelectElement.js	Element selection dialog	hclient/widgets/cms/editCMS_SelectElement.js

Sources:

hclient/widgets/cms/editCMS2.js

42-125

hclient/widgets/cms/editCMS_WidgetCfg.js

24-58

hclient/widgets/cms/editCMS_ElementCfg.js

26-45

Widget System

The CMS includes a range of widgets that can be added to pages:

Widget Type	Purpose	Configuration Options
Search Input	Simple search interface	Filters, placeholders, domains
Result List	Display search results	View modes, styling, actions
Map	Geographic visualization	Layers, controls, popups
Story Map	Narrative geographic presentation	Story format, element ordering
Navigation	Menu creation	Orientation, styling
Email Form	Contact functionality	Recipient, fields, styling

Sources:

hclient/widgets/cms/editCMS_WidgetCfg.html

54-1127

Multi-language Support

The CMS supports multiple languages for content:

```
// From websiteRecord.php - detecting language preferences
$website_languages = @$rec['details'][DT_LANGUAGES];
if(!isEmptyArray($website_languages)){
    $website_languages_codes = getTermCodes($mysqli, $website_languages);
    foreach($website_languages as $term_id){
        $lang_code = @$website_languages_codes[$term_id];
        if($lang_code){
```

```
$lang_code = strtoupper($lang_code);  
    // Process Language options  
    }  
}  
}
```

Sources: `hclient/widgets/cms/websiteRecord.php` 248-271

`hclient/widgets/cms/websiteScriptAndStyles.php` 442-467

Custom Styling and Scripting

The CMS allows custom CSS and JavaScript to be added to websites:

```
// From websiteScriptAndStyles.php - handling custom JavaScript  
if(isJsAllowed){  
    website_custom_javascript = __getValue($rec, DT_CMS_SCRIPT);  
}  
//custom styles - mainly to override positions/visibility for #main-xxx elements  
website_custom_css = __getValue($rec, DT_CMS_CSS);
```

Sources: `hclient/widgets/cms/websiteRecord.php` 305-321

`hclient/widgets/cms/editCMS_ElementCfg.js` 285-339

Data Analysis and Visualization

Heurist provides specialized visualization tools for analyzing and presenting data in compelling ways.

Story Maps

Story maps combine geographic visualization with narrative content, allowing users to create guided journeys through spatiotemporal data:

Sources: `hclient/widgets/viewers/app_storymap.js` 23-85

Key features of Story Maps include:

- Linking narrative content with geographic locations
- Sequential progression through story elements

- Temporal filtering and animation
- Custom styling for story elements
- Different display modes (vertical, slide, tabs)

Sources: `hclient/widgets/viewers/app_storymap.js` 114-174

`hclient/widgets/cms/editCMS_WidgetCfg.html` 215-282

Time Maps

Time Maps integrate mapping with timeline visualization to explore data across both space and time:

```
// From app_timemap.js - handling search results
_handleSearchEvent: function(e, data) {
    // Handle search events and update map accordingly
    if(data.search_realm && data.search_realm==this.options.search_realm){
        this.recordset_changed = true;
        this.option("recordset", data.recordset);
        this._refresh();
    }
}
```

Sources: `hclient/widgets/viewers/app_timemap.js` 135-210

Key features of Time Maps include:

- Synchronized map and timeline views
- Filtering by time period
- Animation of temporal data
- Integration with search system
- Support for multiple map documents

Sources: `hclient/widgets/viewers/app_timemap.js` 456-510

Extended Result List

The Record List Extensions allow for more advanced display of search results, including:

- Integration with external smarty templates
- Custom CSS styling
- Selection-based filtering

- Print and export functionality
- Integration with external applications

```
// From recordListExt.js - handling record selection
_onRecordSelection: function(event, data){
  if(data && data.selection){
    this.options.selection = data.selection;
    if(this.options.is_single_selection || this.options.is_multi_selection){
      this._refresh();
    }
  }
}
```

Sources: `hclient/widgets/viewers/recordListExt.js` 23-63

`hclient/widgets/viewers/recordListExt.js` 336-386

Integration Between Advanced Features

The advanced features can be combined to create sophisticated applications:

Integration	Description	Implementation
CMS + Mapping	Interactive maps embedded in websites	Widget configuration
CMS + Story Maps	Narrative geographic experiences	Widget configuration
Mapping + Data Analysis	Geographic visualization of analytical results	Search integration
Time Maps + Result List	Temporospatial filtering with detailed results	Search realm connection

Sources: `hclient/widgets/cms/editCMS_WidgetCfg.js` 197-208

`hclient/widgets/viewers/app_timemap.js` 368-413

Configuration Options

Each advanced feature offers extensive configuration options:

Map Configuration

The mapping system can be configured with:

- Base map selection
- Layer visibility and styling

- Control panel options
- Temporal settings
- Popup formatting
- Drawing tools permissions

Sources: `viewers/map/mapping.js` 118-155

CMS Configuration

Website generation can be configured with:

- Template selection
- Header and footer content
- Menu structure
- Widget positioning and styling
- Multi-language content
- Custom JavaScript and CSS

Sources: `hclient/widgets/cms/websiteRecord.php` 407-420

`hclient/widgets/cms/editCMS_WidgetCfg.html` 54-1127

Visualization Configuration

Story Maps and Time Maps can be configured with:

- Display modes
- Element ordering
- Animation settings
- Styling options
- Interaction behaviors

Sources: `hclient/widgets/viewers/app_storymap.js` 26-83

`hclient/widgets/viewers/app_timemap.js` 27-57

Summary

Heurist's advanced features provide powerful tools for geographic visualization, website generation, and data analysis that extend the core functionality of the system. These features are

highly configurable and can be combined to create sophisticated applications for exploring, analyzing, and presenting research data.

Sources: `viewers/map/mapping.js` `hclient/widgets/cms/websiteRecord.php`